

NUMBER PATTERNS

Short Revision Notes · Grade 9 Mathematics

Apply Level — Use Knowledge in New Situations

1. USING THE GENERAL TERM FORMULA

$$T_n = \text{First term} + (n - 1) \times \text{common difference}$$

$$\text{OR: } T_n = dn + c \text{ (d = common difference, c = constant)}$$

2. FINDING SPECIFIC TERMS

Task	Method	Example
Find T_1, T_2, T_3	Substitute $n = 1, 2, 3$	$T_n = 3n + 2 \rightarrow T_1 = 5, T_2 = 8, T_3 = 11$
Find T_{20}	Substitute $n = 20$	$T_n = 7n + 1 \rightarrow T_{20} = 7(20) + 1 = 141$
Find T_{12}	Substitute $n = 12$	$T_n = 50 - 7n \rightarrow T_{12} = 50 - 84 = -34$
Find T_{100}	Substitute $n = 100$	$T_n = 4n - 3 \rightarrow T_{100} = 400 - 3 = 397$

3. FINDING THE TERM POSITION

Task	Method	Example
Which term equals a value?	Set $T_n = \text{value}$, solve for n	$T_n = 4n - 3 = 97 \rightarrow 4n = 100 \rightarrow n = 25$ (25th term)
Is a value a term?	Solve for n . If n is a whole number $\rightarrow$ YES	$T_n = 56 - 4n = 0 \rightarrow n = 14 \rightarrow$ YES (14th term)
Is a value NOT a term?	Solve for n . If n is NOT a whole number $\rightarrow$ NO	$T_n = 56 - 4n = 18 \rightarrow n = 9.5 \rightarrow$ NO

4. DEVELOPING THE GENERAL TERM

Step	Action	Example: 5, 11, 17, 23
1	Find the common difference (d)	$11 - 5 = 6$
2	Write the form: $T_n = dn + c$	$T_n = 6n + c$
3	Substitute $n = 1$ and $T_1 = 5$	$5 = 6(1) + c \rightarrow c = -1$
4	Write the general term	$T_n = 6n - 1$

Alternative method (first term formula):

$$T_n = \text{First term} + (n-1) \times d$$

$$T_n = 5 + (n-1) \times 6$$

$$T_n = 5 + 6n - 6$$

$$T_n = 6n - 1$$

5. APPLY TO DIFFERENT SEQUENCES

Sequence	Common Difference (d)	General Term (T_n)
5, 8, 11, 14, ...	3	$3n + 2$
1, 7, 13, 19, ...	6	$6n - 5$
9, 17, 25, 33, ...	8	$8n + 1$
4, 10, 16, 22, ...	6	$6n - 2$
22, 19, 16, 13, ...	-3	$25 - 3n$
22, 20, 18, 16, ...	-2	$24 - 2n$
1, 4, 7, 10, ...	3	$3n - 2$
10, 17, 24, 31, ...	7	$7n + 3$

6. FINDING (n+1)th TERM

Task	Method	Example ($T_n = 2n + 3$)
Find T_{n+1}	Substitute (n+1) for n	$T_{n+1} = 2(n+1) + 3 = 2n + 5$
Find T_{n-1}	Substitute (n-1) for n	$T_{n-1} = 2(n-1) + 3 = 2n + 1$
Find T_{2n}	Substitute 2n for n	$T_{2n} = 2(2n) + 3 = 4n + 3$

7. SOLVING WORD PROBLEMS

Step	Action	Example
1	Identify the pattern	A runner runs 500m on day 1, 600m on day 2, 700m on day 3...
2	Find common difference	$600 - 500 = 100$ (d = 100)
3	Find first term	$T_1 = 500$
4	Write general term	$T_n = 500 + (n-1) \times 100 = 100n + 400$
5	Substitute for required value	$T_{20} = 100(20) + 400 = 2400$ m
6	Solve for n if needed	$3000 = 100n + 400 \rightarrow n = 26$ (26th day)

8. COMMON WORD PROBLEM TYPES

Type	Example	Solution Method
Distance / Time	A runner increases distance by 100m each day	$T_n = \text{first distance} + (n-1) \times \text{increase}$
Money / Finance	A person saves Rs 100, then Rs 10 each week	$T_n = \text{initial} + (n-1) \times \text{weekly amount}$
Seating / Arrangements	First row 9 seats, second row 12, third row 15...	$T_n = \text{first} + (n-1) \times \text{increase}$
Matchstick Patterns	Figure 1 uses 6 sticks, figure 2 uses 9 sticks...	$T_n = \text{first} + (n-1) \times \text{increase}$

9. APPLYING TO MATCHSTICK PATTERNS

Figure Number	Total Matchsticks	Common Difference	General Term
1	6	3	$T_n = 3n + 3$
2	9	3	$T_n = 3n + 3$
3	12	3	$T_n = 3n + 3$
4	15	3	$T_n = 3n + 3$

Check: $T_1 = 3(1) + 3 = 6$ ✓ $T_2 = 3(2) + 3 = 9$ ✓

10. APPLICATION TIPS

Tip	Explanation	Example
Identify d first	Always find the common difference before anything else	5, 8, 11, 14 $\rightarrow d = 3$
Check your work	Substitute $n = 1, 2, 3$ to verify T_n	$T_n = 3n + 2 \rightarrow T_1 = 5, T_2 = 8, T_3 = 11$ ✓
Whole numbers only	n must be a positive integer	$n = 1, 2, 3, 4, \dots$
Negative d is fine	The formula works with negative numbers	$d = -4 \rightarrow T_n = a + (n-1)(-4)$

11. PRACTICE PROBLEMS

No.	Problem	Solution
1	Find the 15th term: $T_n = 3n - 2$	$T_{15} = 3(15) - 2 = 43$
2	Find the 12th term: $T_n = 5n + 1$	$T_{12} = 5(12) + 1 = 61$
3	Which term is 35? $T_n = 3n + 2$	$35 = 3n + 2 \rightarrow n = 11$
4	Write T_n for: 7, 11, 15, 19, ...	$d = 4 \rightarrow T_n = 4n + 3$
5	Write T_n for: 20, 17, 14, 11, ...	$d = -3 \rightarrow T_n = 23 - 3n$
6	Find T_{20} for: $T_n = 7n - 3$	$T_{20} = 7(20) - 3 = 137$

12. QUICK APPLICATION FORMULAS

To Find	Formula
Any term	$T_n = T_1 + (n-1)d$
Common difference	$d = T_2 - T_1$
General term from sequence	$T_n = T_1 + (n-1)d$
Term position	$n = (T_n - T_1) \div d + 1$
Next term	$T_{n+1} = T_n + d$

13. SELF-TEST — APPLICATION QUESTIONS

No.	Question	Answer
1	$T_n = 4n - 7$. Find T_{10}	33
2	Find the 20th term of 5, 8, 11, 14, ...	62
3	Which term of $T_n = 5n - 2$ is 48?	10th term
4	$T_n = 50 - 6n$. Find T_8	2
5	Write T_n for: 10, 13, 16, 19, ...	$3n + 7$
6	Write T_n for: 30, 27, 24, 21, ...	$33 - 3n$
7	A runner runs 400m on day 1, adds 50m daily. How far on day 10?	850 m
8	First row has 10 seats, adds 4 seats per row. How many in row 15?	66 seats
9	$T_n = 7n + 5$. Find T_{25}	180
10	Which term is 100 in $T_n = 4n + 8$?	23rd term

★ **Revision Tip:** Practice applying the formula to different scenarios. Start with simple number sequences, then move to word problems like matchstick patterns and real-life situations!